CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

FOREBAY FARMS- ROAD 6 DAIRY

5615 AVE 24

CHOWCHILLA CA 93610

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Forebay Farms LLC - Road 6

Physical address of dairy:

Number and Street	City	County	Zip Code
5615 Ave 24	Chowchilla	Madera	93610

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X020-X012-X002-XXXX X020-X012-X003-XXXX

B. OPERATORS

Fagundes, Lloyd

Operator name: Fagundes, Lloyd

Telephone no.: Landline: ; Cellular: (209) 383-6075

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

Forebay Farms

Operator name: Forebay Farms

Telephone no.: Landline: ; Cellular: (209) 383-6075

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This operator is responsible for paying permit fees.

C. OWNERS

Fagundes, Lloyd

Legal owner name: Fagundes, Lloyd

Telephone no.: Landline: ; Cellular: (209) 383-6075

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

Forebay Farms

Legal owner name:	Forebay Farms	Telephone no.:	Landline	(209) 383-6075 Cellular
P.O. Box 2717	Merced	CA	95344	
Mailing Address Number and Street	City	State	Zip Code	

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 2 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 51,093,000 *gallons*

Total nitrogen generated: 7,594.00 *lbs*

Total phosphorus generated: 3,831.87 *lbs*

Total potassium generated: 50,628.60 *lbs*

Total salt generated: 253,865.34 *lbs*

	51,093,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	51,093,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	Corral solids *Solid Manure Imported*	2,860.00 *ton*	Dry-weight	60.7	10,200.00	12,300.00	4,600.00		0.00

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	22,929.19	27,649.91	10,340.62	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	22,929.19	27,649.91	10,340.62	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	45	45	2	both	X020-X012-X003-XXXX
2	26	26	2	both	X020-X012-X002-XXXX
3	40	40	2	both	X020-X040-X002-XXXX
4	86	86	2	both	X020-X040-X005-XXXX
Totals for areas that were used for application	197	197	8		
Totals for areas that were not used for application					
Land application area totals	197	197	8		

B. CROPS AND HARVESTS

1

Field name: 1

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	747.00 *ton*	Dry-weight		21.3	12,850.00	3,350.00	21,750.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.60	335.75	87.53	568.29	2,142.53

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	979.00 *ton*	Dry-weight		63.9	11,400.00	2,700.00	13,100.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.76	179.07	42.41	205.77	911.04

2

Field name: 2

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	446.00 *ton*	Dry-weight		30.7	11,750.00	3,200.00	21,500.00		8.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.15	279.36	76.08	511.17	1,937.68

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	565.00 *ton*	Dry-weight		67.0	12,900.00	1,900.00	15,100.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.73	185.02	27.25	216.57	946.59

3

Field name: 3

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	721.00 *ton*	Dry-weight		63.4	12,000.00	3,200.00	21,300.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.03	158.33	42.22	281.04	1,029.16

3

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	870.00 *ton*	Dry-weight		62.4	10,500.00	2,200.00	10,400.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.75	171.74	35.98	170.10	965.00

4

Field name: 4

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 86 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	1,487.00 *ton*	Dry-weight		33.1	12,200.00	3,400.00	22,300.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.29	282.25	78.66	515.91	1,920.20

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 86 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	1,870.00 *ton*	Dry-weight		68.3	13,100.00	2,400.00	11,700.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.74	180.59	33.09	161.29	785.79

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/20/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough — Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	9.53	3.22	24.43	199.78	1,890,000.00 *gal*
Irrigation Well	Ground water	6.89	0.00	0.00	0.00	6,630,000.00 *gal*
Application event totals		16.42	3.22	24.43	199.78	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	8.76	5.22	79.21	357.50	1,890,000.00 *gal*
Irrigation Well	Ground water	6.28	0.00	0.00	0.00	6,045,000.00 *gal*
Application event totals		15.04	5.22	79.21	357.50	

1 - 06/11/2023: Corn, silage

Field name: 1

Crop: Corn, silage — Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported	Corral solids	116.34	140.29	52.47	0.00	653.00 *ton*
Application event totals		116.34	140.29	52.47	0.00	

1 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.30	4.35	66.01	297.92	1,575,000.00 *gal*
Irrigation Well	Ground water	7.78	0.00	0.00	0.00	7,488,000.00 *gal*
Application event totals		15.08	4.35	66.01	297.92	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.30	4.35	66.01	297.92	1,575,000.00 *gal*
Irrigation Well	Ground water	7.70	0.00	0.00	0.00	7,410,000.00 *gal*
Application event totals		15.00	4.35	66.01	297.92	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.87	0.76	7.01	44.69	1,575,000.00 *gal*
Irrigation Well	Ground water	8.83	0.00	0.00	0.00	8,502,000.00 *gal*
Application event totals		10.70	0.76	7.01	44.69	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.07	0.00	0.00	0.00	8,736,000.00 *gal*
Application event totals		9.07	0.00	0.00	0.00	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.87	0.76	7.01	44.69	1,575,000.00 *gal*
Irrigation Well	Ground water	8.10	0.00	0.00	0.00	7,800,000.00 *gal*
Application event totals		9.97	0.76	7.01	44.69	

1 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.87	0.76	7.01	44.69	1,575,000.00 *gal*
Irrigation Well	Ground water	7.45	0.00	0.00	0.00	7,176,000.00 *gal*
Application event totals		9.32	0.76	7.01	44.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.86	0.00	0.00	0.00	7,566,000.00 *gal*
Application event totals		7.86	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.21	0.00	0.00	0.00	6,942,000.00 *gal*
Application event totals		7.21	0.00	0.00	0.00	

2 - 10/20/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	9.53	3.22	24.43	199.78	1,092,000.00 *gal*
Irrigation Well	Ground water	5.56	0.00	0.00	0.00	4,440,000.00 *gal*
Application event totals		15.09	3.22	24.43	199.78	

2 - 10/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	8.76	5.22	79.21	357.50	1,092,000.00 *gal*
Irrigation Well	Ground water	3.76	0.00	0.00	0.00	3,000,000.00 *gal*
Application event totals		12.52	5.22	79.21	357.50	

2 - 06/11/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported	Corral solids	116.56	140.55	52.57	0.00	378.00 *ton*
Application event totals		116.56	140.55	52.57	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.41	4.42	67.02	302.50	924,000.00 *gal*
Irrigation Well	Ground water	5.56	0.00	0.00	0.00	4,440,000.00 *gal*
Application event totals		12.97	4.42	67.02	302.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.41	4.42	67.02	302.50	924,000.00 *gal*
Irrigation Well	Ground water	5.11	0.00	0.00	0.00	4,080,000.00 *gal*
Application event totals		12.52	4.42	67.02	302.50	

2 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.90	0.77	7.12	45.37	924,000.00 *gal*
Irrigation Well	Ground water	6.61	0.00	0.00	0.00	5,280,000.00 *gal*
Application event totals		8.51	0.77	7.12	45.37	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.31	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		6.31	0.00	0.00	0.00	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.90	0.77	7.12	45.37	924,000.00 *gal*
Irrigation Well	Ground water	5.26	0.00	0.00	0.00	4,200,000.00 *gal*
Application event totals		7.16	0.77	7.12	45.37	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.90	0.77	7.12	45.37	924,000.00 *gal*
Irrigation Well	Ground water	6.16	0.00	0.00	0.00	4,920,000.00 *gal*
Application event totals		8.06	0.77	7.12	45.37	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.31	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		6.31	0.00	0.00	0.00	

2 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.51	0.00	0.00	0.00	3,600,000.00 *gal*
Application event totals		4.51	0.00	0.00	0.00	

3 - 10/24/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	9.53	3.22	24.43	199.78	1,680,000.00 *gal*
Irrigation Well	Ground water	6.88	0.00	0.00	0.00	5,889,000.00 *gal*
Application event totals		16.41	3.22	24.43	199.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	8.76	5.22	79.21	357.50	1,680,000.00 *gal*
Irrigation Well	Ground water	6.29	0.00	0.00	0.00	5,382,000.00 *gal*
Application event totals		15.05	5.22	79.21	357.50	

3 - 06/11/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported	Corral solids	116.45	140.42	52.52	0.00	581.00 *ton*
Application event totals		116.45	140.42	52.52	0.00	

05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.34	4.37	66.34	299.41	1,407,000.00 *gal*
Irrigation Well	Ground water	7.75	0.00	0.00	0.00	6,630,000.00 *gal*
Application event totals		15.08	4.37	66.34	299.41	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.34	4.37	66.34	299.41	1,407,000.00 *gal*
Irrigation Well	Ground water	5.36	0.00	0.00	0.00	6,591,000.00 *gal*
Application event totals		12.70	4.37	66.34	299.41	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.88	0.76	7.04	44.91	1,407,000.00 *gal*
Irrigation Well	Ground water	8.84	0.00	0.00	0.00	7,566,000.00 *gal*
Application event totals		10.72	0.76	7.04	44.91	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.07	0.00	0.00	0.00	7,761,000.00 *gal*
Application event totals		9.07	0.00	0.00	0.00	

3 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.88	0.76	7.04	44.91	1,407,000.00 *gal*
Irrigation Well	Ground water	8.11	0.00	0.00	0.00	6,942,000.00 *gal*
Application event totals		9.99	0.76	7.04	44.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.88	0.76	7.04	44.91	1,407,000.00 *gal*
Irrigation Well	Ground water	7.47	0.00	0.00	0.00	6,396,000.00 *gal*
Application event totals		9.35	0.76	7.04	44.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.84	0.00	0.00	0.00	6,708,000.00 *gal*
Application event totals		7.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.20	0.00	0.00	0.00	6,162,000.00 *gal*
Application event totals		7.20	0.00	0.00	0.00	

4 - 10/26/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	9.53	3.22	24.43	199.78	3,612,000.00 *gal*
Irrigation Well	Ground water	5.56	0.00	0.00	0.00	14,687,520.00 *gal*
Application event totals		15.09	3.22	24.43	199.78	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	8.76	5.22	79.21	357.50	3,612,000.00 *gal*
Irrigation Well	Ground water	3.76	0.00	0.00	0.00	9,924,000.00 *gal*
Application event totals		12.52	5.22	79.21	357.50	

4 - 06/11/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported	Corral solids	116.44	140.41	52.51	0.00	1,249.00 *ton*
Application event totals		116.44	140.41	52.51	0.00	

05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.28	4.34	65.86	297.22	3,003,000.00 *gal*
Irrigation Well	Ground water	5.56	0.00	0.00	0.00	14,687,520.00 *gal*
Application event totals		12.84	4.34	65.86	297.22	

4 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	7.28	4.34	65.86	297.22	3,003,000.00 *gal*
Irrigation Well	Ground water	5.11	0.00	0.00	0.00	13,496,640.00 *gal*
Application event totals		12.39	4.34	65.86	297.22	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.86	0.76	6.99	44.58	3,003,000.00 *gal*
Irrigation Well	Ground water	6.61	0.00	0.00	0.00	17,466,240.00 *gal*
Application event totals		8.47	0.76	6.99	44.58	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.31	0.00	0.00	0.00	16,672,320.00 *gal*
Application event totals		6.31	0.00	0.00	0.00	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.86	0.76	6.99	44.58	3,003,000.00 *gal*
Irrigation Well	Ground water	5.26	0.00	0.00	0.00	13,893,600.00 *gal*
Application event totals		7.12	0.76	6.99	44.58	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pond Water	Process wastewater	1.86	0.76	6.99	44.58	3,003,000.00 *gal*
Irrigation Well	Ground water	6.16	0.00	0.00	0.00	16,275,360.00 *gal*
Application event totals		8.02	0.76	6.99	44.58	

4 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.31	0.00	0.00	0.00	16,672,320.00 *gal*
Application event totals		6.31	0.00	0.00	0.00	

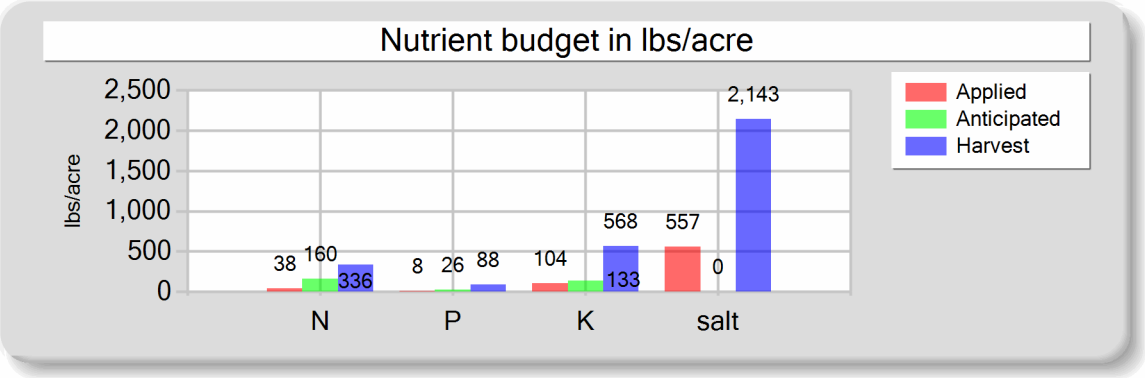
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.51	0.00	0.00	0.00	11,908,800.00 *gal*
Application event totals		4.51	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 10/20/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	18.30	8.45	103.64	557.28
Fresh water	13.16	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	38.46	8.45	103.64	557.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	335.75	87.53	568.29	2,142.53
Nutrient balance	-297.29	-79.08	-464.65	-1,585.25
Applied to removed ratio	0.11	0.10	0.18	0.26

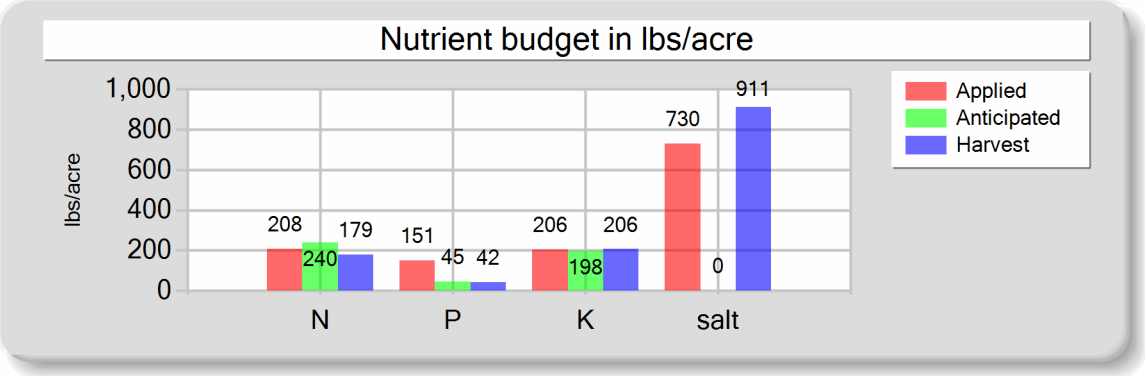
Fresh water applied	
12,675,000.00	*gallons*
466.78	*acre-inches*
10.37	*inches/acre*

Process wastewater applied	
3,780,000.00	*gallons*
139.20	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/11/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	116.34	140.29	52.47	0.00
Process wastewater	20.21	10.98	153.05	729.90
Fresh water	63.99	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.54	151.27	205.51	729.90
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	179.07	42.41	205.77	911.04
Nutrient balance	28.48	108.86	-0.25	-181.14
Applied to removed ratio	1.16	3.57	1.00	0.80

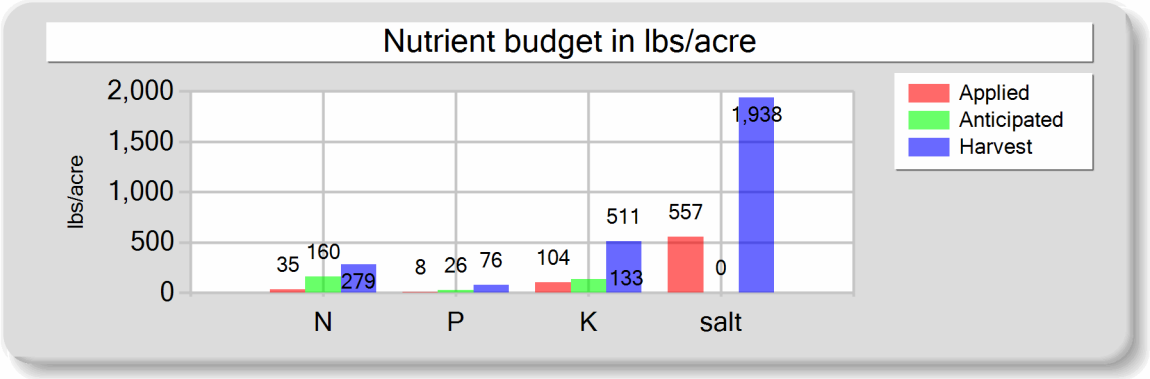
Fresh water applied	
61,620,000.00	*gallons*
2,269.26	*acre-inches*
50.43	*inches/acre*

Process wastewater applied	
7,875,000.00	*gallons*
290.01	*acre-inches*
6.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/20/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	18.30	8.45	103.64	557.28
Fresh water	9.31	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	34.61	8.45	103.64	557.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	279.36	76.08	511.17	1,937.68
Nutrient balance	-244.75	-67.63	-407.53	-1,380.40
Applied to removed ratio	0.12	0.11	0.20	0.29

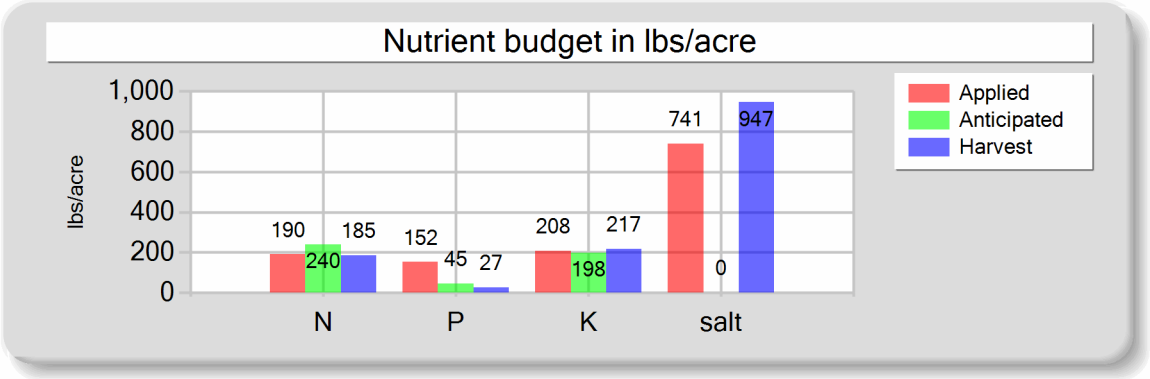
Fresh water applied	
7,440,000.00	*gallons*
273.99	*acre-inches*
10.54	*inches/acre*

Process wastewater applied	
2,184,000.00	*gallons*
80.43	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/11/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	116.56	140.55	52.57	0.00
Process wastewater	20.52	11.15	155.40	741.12
Fresh water	45.81	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	189.89	151.71	207.97	741.12
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	185.02	27.25	216.57	946.59
Nutrient balance	4.88	124.46	-8.60	-205.47
Applied to removed ratio	1.03	5.57	0.96	0.78

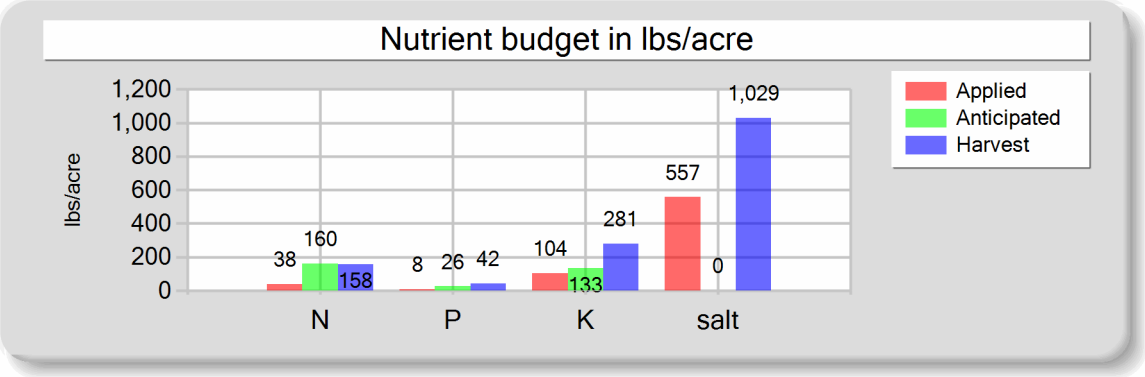
Fresh water applied	
36,600,000.00	*gallons*
1,347.85	*acre-inches*
51.84	*inches/acre*

Process wastewater applied	
4,620,000.00	*gallons*
170.14	*acre-inches*
6.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/24/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	18.30	8.45	103.64	557.28
Fresh water	13.17	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	38.46	8.45	103.64	557.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	158.33	42.22	281.04	1,029.16
Nutrient balance	-119.87	-33.77	-177.40	-471.88
Applied to removed ratio	0.24	0.20	0.37	0.54

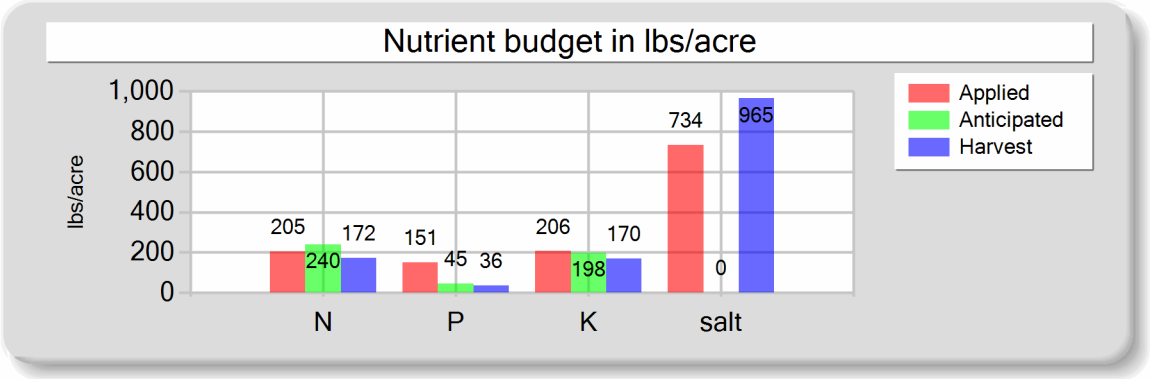
Fresh water applied	
11,271,000.00	*gallons*
415.07	*acre-inches*
10.38	*inches/acre*

Process wastewater applied	
3,360,000.00	*gallons*
123.74	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/11/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	116.45	140.42	52.52	0.00
Process wastewater	20.31	11.04	153.81	733.54
Fresh water	61.63	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.40	151.46	206.33	733.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	171.74	35.98	170.10	965.00
Nutrient balance	33.66	115.48	36.23	-231.46
Applied to removed ratio	1.20	4.21	1.21	0.76

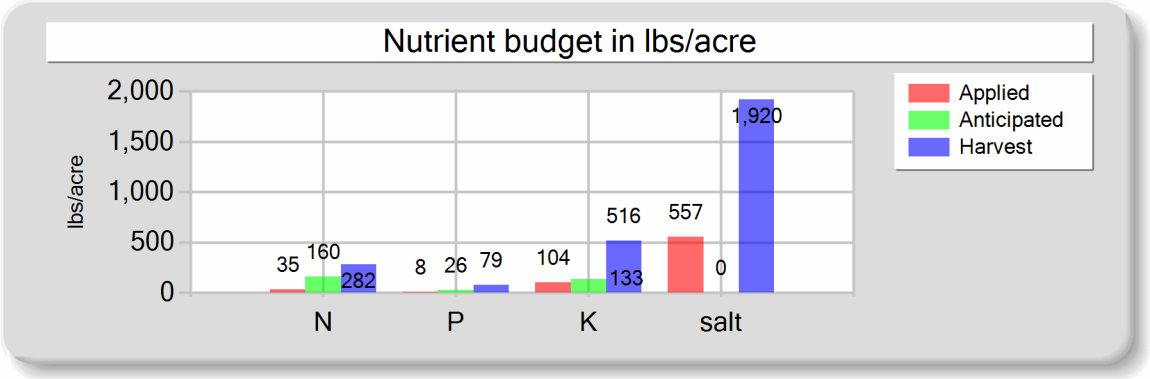
Fresh water applied	
54,756,000.00	*gallons*
2,016.48	*acre-inches*
50.41	*inches/acre*

Process wastewater applied	
7,035,000.00	*gallons*
259.08	*acre-inches*
6.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/26/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	18.30	8.45	103.64	557.28
Fresh water	9.31	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	34.61	8.45	103.64	557.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	282.25	78.66	515.91	1,920.20
Nutrient balance	-247.64	-70.21	-412.27	-1,362.92
Applied to removed ratio	0.12	0.11	0.20	0.29

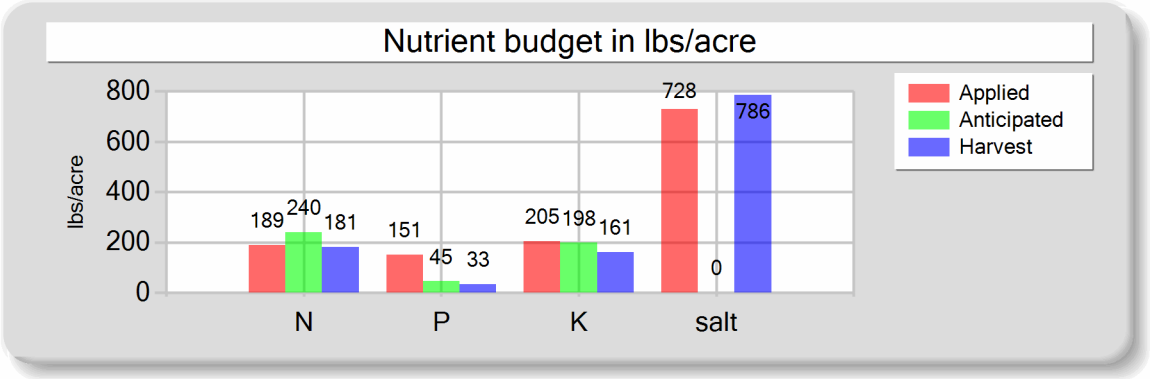
Fresh water applied	
24,611,520.00	*gallons*
906.36	*acre-inches*
10.54	*inches/acre*

Process wastewater applied	
7,224,000.00	*gallons*
266.04	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/11/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	116.44	140.41	52.51	0.00
Process wastewater	20.16	10.96	152.69	728.20
Fresh water	45.82	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	189.42	151.36	205.20	728.20
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	180.59	33.09	161.29	785.79
Nutrient balance	8.82	118.28	43.91	-57.59
Applied to removed ratio	1.05	4.57	1.27	0.93

Fresh water applied	
121,072,800.00	*gallons*
4,458.70	*acre-inches*
51.85	*inches/acre*

Process wastewater applied	
15,015,000.00	*gallons*
552.95	*acre-inches*
6.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/30/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	26.80	23.60	0.00	0.40	9.20	69.70								1,130.00	570
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	42.50	30.40	0.00	1.90	26.70	184.00								1,550.00	900
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	20.30	1.10	0.00	4.70	14.90	226.00								1,550.00	1,020
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.10	0.56	0.00	0.30	2.60	24.00								228.00	153
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.88	0.00	0.00	2.90	7.05	64.90								548.00	425
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Domestic Well

Domestic Well

Sample description: Domestic Well

Sample date: 08/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.90	0.00	5.90								427.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

New Well Ave 24

Sample description: New Well Ave 24

Sample date: 08/09/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.90	0.00	3.90								412.00	0
DL	0.01	0.01	0.01								10.00	1

IW #8

Sample description: IW #8

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	5.60								427.00	0
DL		0.01	0.01								0.01	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/20/2022: Oats, silage-soft dough

Field 1A Oats

Sample and source description: Field 1A Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 25.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	3,300.00	21,400.00		8.10
DL	0.01	0.01	0.01		1.00

1 - 10/20/2022: Oats, silage-soft dough

Field 1B Oats

Sample and source description: Field 1B Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	3,400.00	22,100.00		8.30
DL	0.01	0.01	0.01		1.00

1 - 06/11/2023: Corn, silage

Corn Field 1

Sample and source description: Corn Field 1

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,700.00	13,100.00		5.80
DL	0.01	0.01	0.01		1.00

2 - 10/20/2022: Oats, silage-soft dough

Field 2A Oats

Sample and source description: Field 2A Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 30.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	3,100.00	21,000.00		8.00
DL	0.01	0.01	0.01		1.00

2 - 10/20/2022: Oats, silage-soft dough

Field 2B Oats

Sample and source description: Field 2B Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	3,300.00	22,000.00		8.30
DL	0.01	0.01	0.01		1.00

2 - 06/11/2023: Corn, silage

Corn Field 2

Sample and source description: Corn Field 2

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	1,900.00	15,100.00		6.60
DL	0.01	0.01	0.01		1.00

3 - 10/24/2022: Oats, silage-soft dough

Field 3A Oats

Sample and source description: Field 3A Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	3,100.00	20,800.00		7.70
DL	0.01	0.01	0.01		1.00

3 - 10/24/2022: Oats, silage-soft dough

Field 3B Oats

Sample and source description: Field 3B Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	3,300.00	21,800.00		7.90
DL	0.01	0.01	0.01		1.00

3 - 06/11/2023: Corn, silage

Corn Field 3

Sample and source description: Corn Field 3

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	2,200.00	10,400.00		5.90
DL	0.01	0.01	0.01		1.00

4 - 10/26/2022: Oats, silage-soft dough

Field 4A Oats

Sample and source description: Field 4A Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 23.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	3,300.00	22,400.00		8.60
DL	0.01	0.01	0.01		1.00

4 - 10/26/2022: Oats, silage-soft dough

Field 4B Oats

Sample and source description: Field 4B Oats

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	3,500.00	22,200.00		8.00
DL	0.01	0.01	0.01		1.00

4 - 06/11/2023: Corn, silage

Corn Field 4

Sample and source description: Corn Field 4

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,400.00	11,700.00		5.70
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

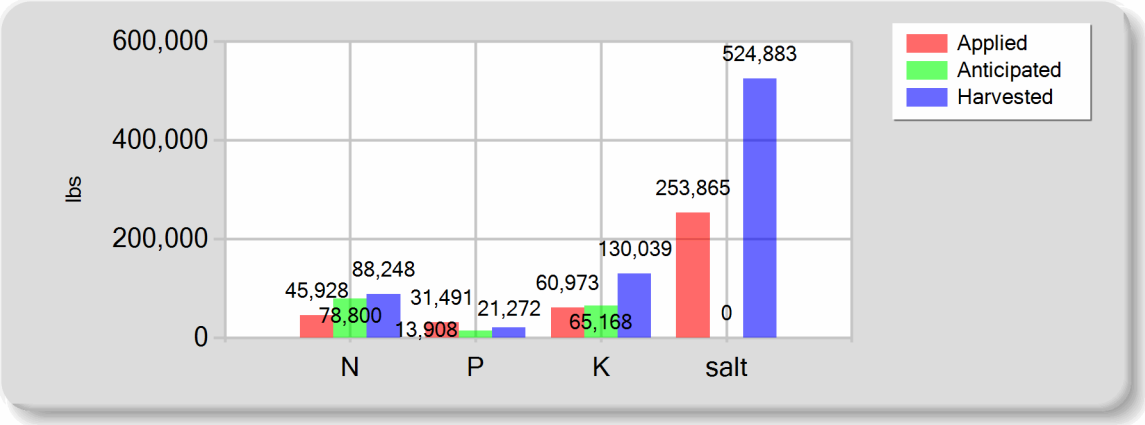
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

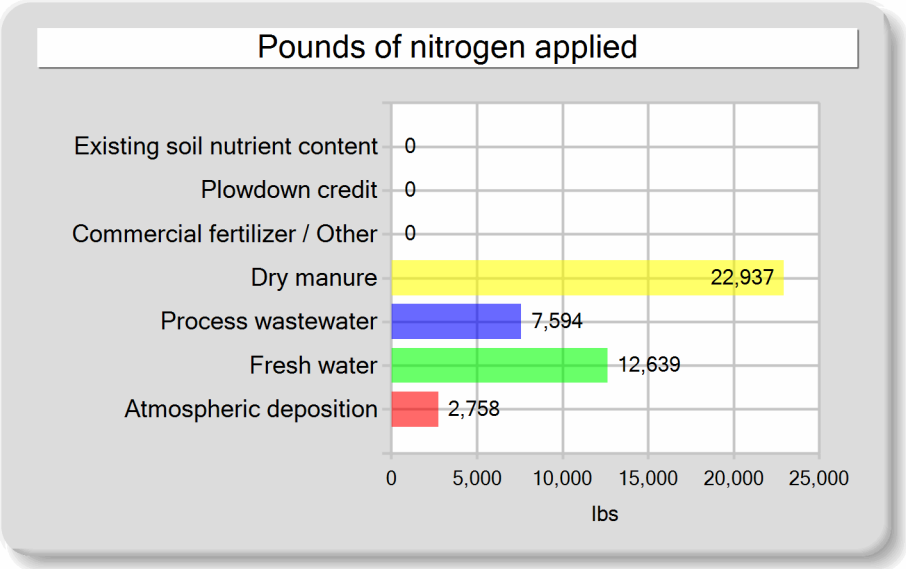
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

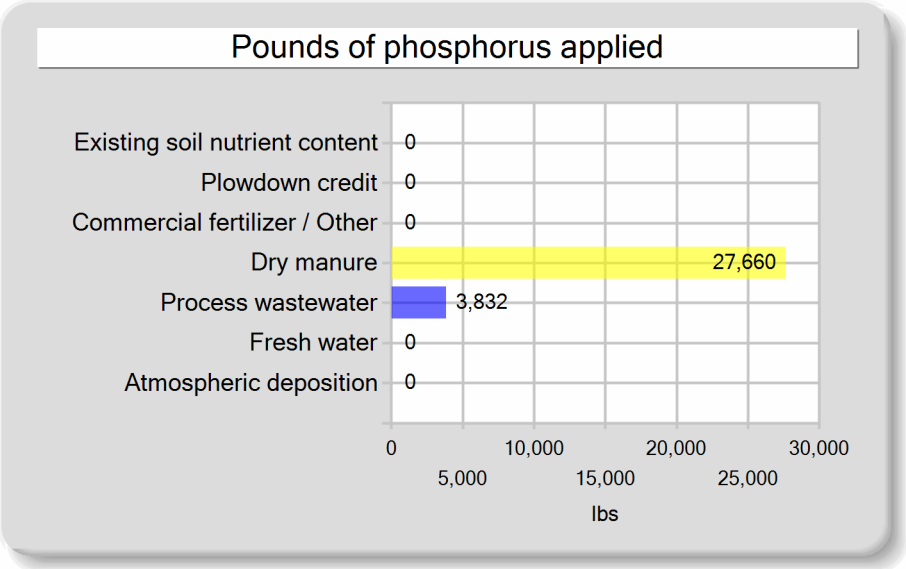
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	22,937.21	27,659.58	10,344.23	0.00
Process wastewater	7,594.00	3,831.87	50,628.60	253,865.34
Fresh water	12,638.70	0.00	0.00	0.00
Atmospheric deposition	2,758.00	0.00	0.00	0.00
Total nutrients applied	45,927.90	31,491.45	60,972.83	253,865.34
Anticipated crop nutrient removal	78,800.00	13,908.20	65,167.60	0.00
Actual crop nutrient removal	88,247.52	21,272.18	130,039.03	524,883.25
Nutrient balance	-42,319.62	10,219.27	-69,066.21	-271,017.91
Applied to removed ratio	0.52	1.48	0.47	0.48

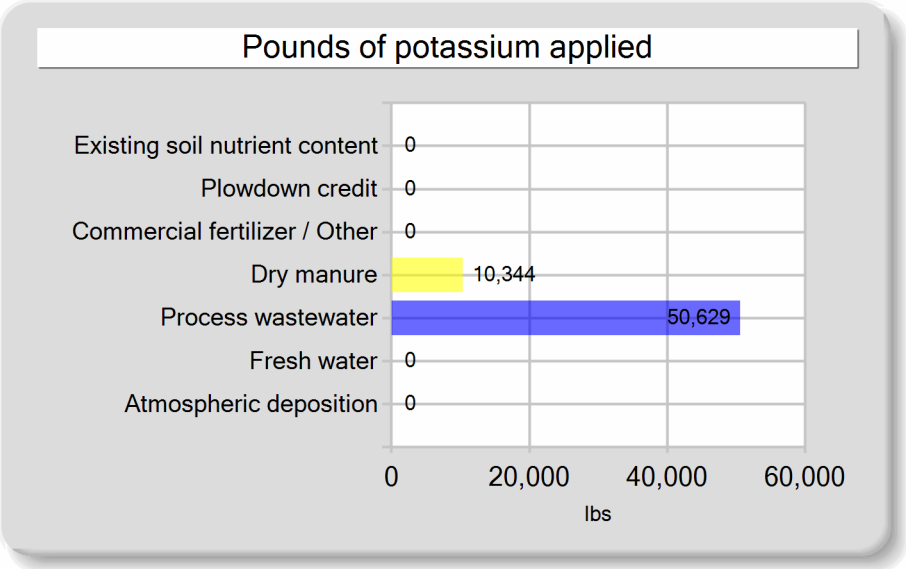
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

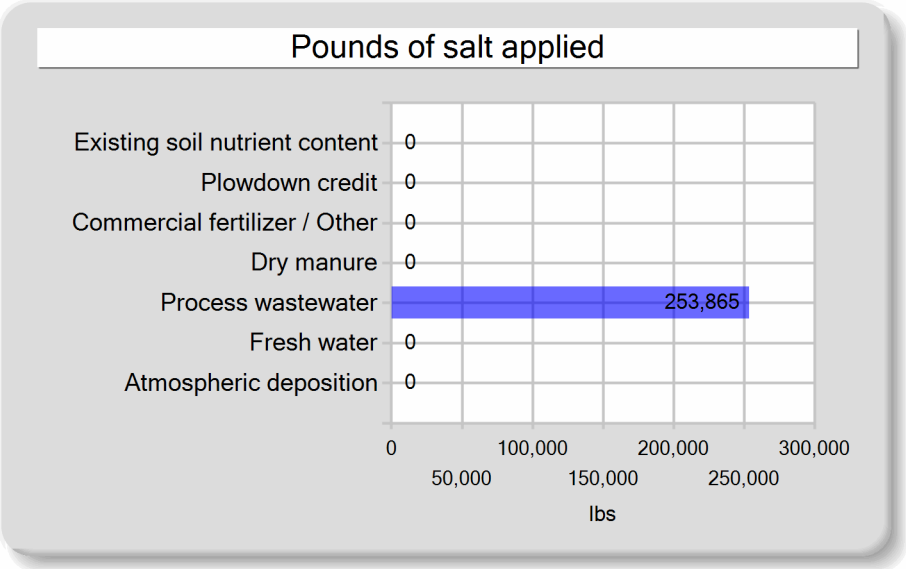


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No animals have been housed onsite since the purchase of this facility, therefore no manure has been produced to allow for sampling, so no manure sample was taken.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Forebay Farms	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23H1169
Sampled By: Tommy
Requested By:
Submitted Date: 08/10/23
Reported Date: 08/18/23
Project:
Crop ID:

Forebay Farms LLC-Road 6
P.O. Box 2717
Merced CA 95344
00-0025035 08

E-mail: amacias@fagundesdairy.com
Copy To: cas.labs@yahoo.cm

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TKN mg/L	Total Nitrogen TN mg/L	pH at 25°C unit
1	New Well Ave 24	8/9/23	15:00	412	0.41	3.9	ND		ND	4.04	7.8

***** = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Forebay Farms LLC-Road 6
P.O. Box 2717
Merced CA 95344
00-0025035 08

E-mail: amacias@fagundesdairy.com
Copy To:

Lab No.: 23H1169
Sampled By: Tommy
Requested By:
Submitted Date: 08/10/23
Reported Date: 08/18/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs *calculated from nitrate-N* lbs/AcFoot	lbs/AcInch
1	New Well Ave 24	8/9/23	15:00	412	0.41	**3.9**	10.65	0.89

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23H1170
Sampled By: Tommy
Requested By: A. Macias
Submitted Date: 08/10/23
Reported Date: 08/21/23
Project:
Crop ID:

Forebay Farms LLC-Road 6
P.O. Box 2717
Merced CA 95344
00-0025035 08

E-mail: amacias@fagundesdairy.com
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	Dom Well	8/10/23	15:00	427	0.43	5.9	ND		8.0

***** = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland

Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Forebay Farms LLC-Road 6
P.O. Box 2717
Merced CA 95344
00-0025035 08

E-mail: amacias@fagundesdairy.com
Copy To: cas.labs@yahoo.com

Lab No.: 23H1170
Sampled By: Tommy
Requested By: A. Macias
Submitted Date: 08/10/23
Reported Date: 08/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs *calculated from nitrate-N* lbs/AcFoot	lbs/AcInch
1	Dom Well	8/10/23	15:00	427	0.43	**5.9**	16.11	1.34

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Forebay Farms LLC-Road 6
P.O. Box 2717
Merced CA 95344
00-0025035 08

E-mail: amacias@fagundesdairy.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L0733
Sampled By: M. Pedroso
Requested By:
Submitted Date: 12/12/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	IW #8	12/11/23	15:40	427	0.43	5.6	ND		8.4

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

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Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Forebay Farms LLC-Road 6
P.O. Box 2717
Merced CA 95344
00-0025035 08

E-mail: amacias@fagundesdairy.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L0733
Sampled By: M. Pedroso
Requested By:
Submitted Date: 12/12/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	IW #8	12/11/23	15:40	427	0.43	**5.6**	15.29	1.27

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.